INCH-POUND

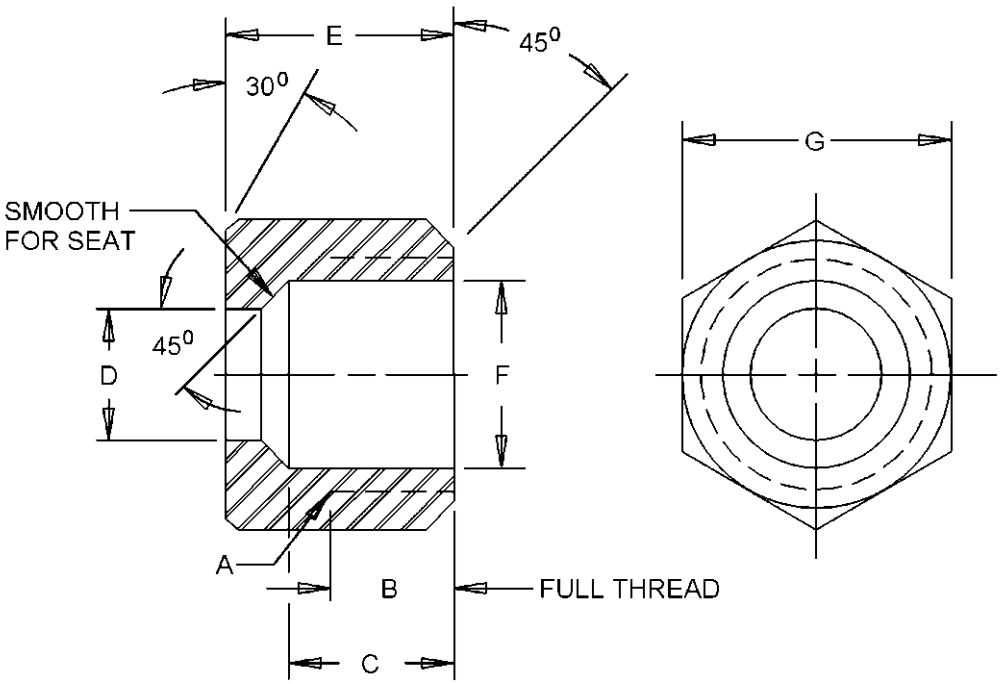
MS39210K
w/ AMENDMENT 1
8 July 2026
SUPERSEDING
MS39210K
31 January 2018

DETAIL SPECIFICATION SHEET

NUTS, TUBE COUPLING – SAFETY SLEEVE COMPRESSION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Nut, tube coupling.

AMSC N/A
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FSC 4730

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Suffix Designator		Tube OD	A Thread (see note 4)	B Thread Length inch (mm)	C inch (mm)
Assembly	Body 6/				
A1	B1	0.1250 (3.175)	0.3125-24UNF-2B	0.219 (5.56)	0.297 (7.54)
A2	B2	0.1875 (4.763)	0.3750-24UNF-2B	0.219 (5.56)	0.297 (7.54)
A3	B3	0.2500 (6.350)	0.5000-20UNF-2B	0.281 (7.14)	0.359 (3.13)
A4	B4	0.3125 (7.938)	0.5625-8UNF-2B	0.281 (7.14)	0.390 (3.92)
A5	B5	0.3750 (9.525)	0.6250-18UNF-2B	0.313 (7.94)	0.422 (10.72)
A6	B6	0.5000 (12.700)	0.7500-16UNF-2B	0.328 (3.33)	0.453 (11.51)
A7	B7	0.6250 (15.875)	0.9375-16UN-2B	0.375 (9.53)	0.469 (11.91)
A8	B8	0.7500 (19.050)	1.0625 16UN-2B	0.406 (10.32)	0.500 (12.70)
A9	B9	0.8750 (22.225)	1.1875-16UN-2B	0.406 (10.32)	0.500 (12.70)
A10	B10	1.0000 (25.400)	1.312516UN-2B	0.406 (10.32)	0.500 (12.70)
A11	B11	1.2500 (31.750)	1.6250-16UN-2B	0.406 (10.32)	0.578 (14.68)
A12	B12	1.5000 (38.100)	1.8750-16UN-2B	0.500 (12.70)	0.672 (17.07)

Suffix Designator		D ± 0.005 (0.13) inch (mm)	E inch (mm)	F ± 0.005 (0.13) inch (mm)	G inch (mm)
Assembly	Body 6/				
A1	B1	0.130 (3.30) 0.134 (3.40)	0.438 (11.11)	0.275 (6.99) 0.270 (6.86)	0.438 (11.11)
A2	B2	0.190 (4.83) 0.194 (4.93)	0.438 (11.11)	0.338 (8.59) 0.333 (8.46)	0.500 (12.70)
A3	B3	0.255 (6.48) 0.259 (6.58)	0.563 (14.29)	0.454 (11.53) 0.449 (11.40)	0.625 (15.88)
A4	B4	0.318 (8.08) 0.322 (8.18)	0.578 (14.68)	0.512 (13.00) 0.505 (12.83)	0.688 (17.46)
A5	B5	0.380 (9.65) 0.384 (9.75)	0.625 (15.88)	0.575 (14.61) 0.568 (14.43)	0.750 (19.05)
A6	B6	0.506 (12.85) 0.510 (12.95)	0.688 (17.46)	0.693 (17.60) 0.686 (17.42)	0.875 (22.23)
A7	B7	0.633 (16.08) 0.637 (16.18)	0.750 (19.05)	0.881 (22.38) 0.874 (22.20)	1.125 (28.58)
A8	B8	0.758 (19.25) 0.762 (19.35)	0.813 (20.64)	1.005 (25.53) 0.999 (25.37)	1.250 (31.75)
A9	B9	0.883 (22.43) 0.887 (22.53)	0.813 (20.64)	1.131 (28.73) 1.124 (28.55)	1.375 (34.93)
A10	B10	1.008 (25.60) 1.012 (25.70)	0.813 (20.64)	1.256 (31.90) 1.249 (31.72)	1.500 (38.10)
A11	B11	1.206 (30.63) 1.264 (32.11)	0.844 (31.43)	1.568 (39.83) 1.561 (39.65)	2.000 (50.80)
A12	B12	1.510 (38.35) 1.514 (38.46)	0.969 (34.61)	1.818 (46.18) 1.810 (45.97)	2.250 (57.15)

FIGURE 1. Nut, tube coupling - Continued.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, the tolerance is ±0.016 (0.41 mm), the tolerance for degrees is ± 2°.
4. Threads shall be in accordance with FED-STD-H28.
5. This illustration is for identification and is not intended to restrict designs or shapes not dimensioned.
6. Assemblies are only furnished to this specification (body, nuts, and sleeves). Bodies are not to be stocked, stored, or issued.

FIGURE 1. Nut, tube coupling - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and table I.

Materials shall be in accordance with MIL-DTL-18866 and table I. Components of the assemblies shall be of the same material and finish.

Working pressure is the established recommended pressure based on a safety factor of 4 to 1, see MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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Table I. Material and finish designators.

Material finish designator	Material	Finish
C	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Carbon steel: Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/
CZ	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 2/
F	Steel	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/
FZ	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 2/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 3/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 3/ 4/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 2/
S	Corrosion resistant steel	Corrosion resistant steel: Passivate in accordance with SAE AMS2700, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 5/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 5/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8 2/
ZC	Steel	Zinc may be any zinc plating from PIN codes CZ, J, and Z with a colored chromate coating. 2/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Not for use in aircraft. Requires approval from the Program Office for all applications.
3/ Hexavalent chromium free. Finish shall be RoHS compliant.
4/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

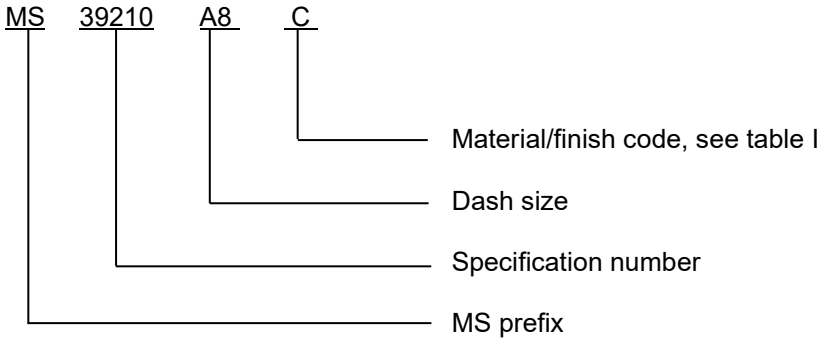
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Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to the primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.

b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Part or Identifying Number (PIN) example:



MS39210A8C indicates a nut for a 0.750 (19.05 mm) inch tube, steel, cadmium plated.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides an alternate material, corrosion resistant steel, and zinc finish via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in MS39210 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

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FED-STD-H28
MIL-DTL-16232
ASTM A29/A29M
ASTM B117
ASTM B633
ASTM B695
ASTM F1136/F1136M
SAE-AMS-QQ-P-416
SAE-AMS-STD-595/36076
SAE-AMS-STD-595/36081
SAE-AMS-STD-595/36099
SAE-AMS-STD-595/36118
SAE-AMS-STD-595/36134
SAE-AMS-STD-595/36152
SAE-AMS-STD-595/36170
SAE-AMS-STD-595/36173
SAE-AMS-STD-595/36176
SAE-AMS-STD-595/36231
SAE-AMS-STD-595/36251
SAE-AMS-STD-595/36270
SAE-AMS-STD-595/36280
SAE-AMS-STD-595/36293
SAE-AMS2417
SAE-AMS2451/5
SAE-AMS2451/9
SAE-AMS2486
SAE-AMS2488
SAE-AMS2700
T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - YD
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2026-045)

Review activities:
Army - AR
Navy - MC, SA

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